

The δJ_7 Principle: Symmetry-Complement Source Sectors in Altermagnetic MnF_2

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A Hamiltonian coefficient can be small in the total energy while remaining leading in a response channel forbidden by a more symmetric reference model. We formalize this distinction for altermagnetic MnF_2 without assuming that the lower-symmetry Hamiltonian is controlled by a single odd parameter. For a specified reference Hamiltonian H_0 , symmetry group G_0 , and finite operator space, group averaging defines an invariant sector and a symmetry-complement sector. Writing the latter as a coordinate vector λ , a symmetry operation acts as $\lambda \mapsto \rho_s \lambda$. An observable transforming with character $\chi_O(s)$ obeys the exact covariance relation $F_O(\rho_s \lambda) = \chi_O(s) F_O(\lambda)$. If $\rho_s = -I$ on the relevant source block, an odd response contains only terms of odd total degree, $F_-(\lambda) = \mathbf{k} \cdot \lambda + O(\|\lambda\|^3)$. The familiar scalar relation $F_-(\lambda) = \chi_1 \lambda + \chi_3 \lambda^3 + \dots$ follows only on a controlled one-dimensional slice for which other co-transforming odd coordinates vanish or are otherwise independently constrained. In MnF_2 , the inequivalence of the two seventh-neighbor Heisenberg bond families provides such a physically motivated slice within the stated range-seven model. With the bond convention fixed explicitly, $\delta J_7 = J_{7b} - J_{7a}$ multiplies a bond-odd operator in the symmetry complement. Published fits and first-principles mappings place $|\delta J_7|$ at only a few μeV , while polarized-neutron modeling identifies this imbalance as the sole two-spin interaction through seventh neighbor that produces a nonvanishing chiral term at quadratic spin-wave order. The result is a conditional source-selection principle, not a universal inverse-amplification law or a proof of absolute microscopic uniqueness. We state the model, slice, and falsification conditions required to attribute a symmetry-selective response to weak exchange.

I. INTRODUCTION

Altermagnets are compensated collinear magnets in which spin and crystal symmetries permit momentum-dependent spin splitting without a net ferromagnetic moment.[1, 2] MnF_2 , historically a canonical easy-axis antiferromagnet, is now a particularly useful test case because several physically distinct small scales coexist in an otherwise simple magnetic system. Unpolarized inelastic neutron scattering found no resolvable altermagnetic exchange splitting within the sensitivity of one experiment,[5] while subsequent polarized-neutron measurements resolved a dipolar-dominated magnon doublet and separately isolated a domain-dependent chiral signal.[6] More recent first-principles work has proposed signed field readouts for the weak exchange imbalance,[10] and polarized-neutron diffraction has added a real-space multipolar view of the ordered magnetization density.[9]

These results expose a question that is broader than MnF_2 : *when can a coefficient that is perturbatively small in the Hamiltonian be the leading coordinate of a selected observable?* A magnitude hierarchy alone cannot answer this. The largest Hamiltonian terms generally control bandwidths, ordering temperatures, and dominant spectral weight, but an observable can be forbidden by the symmetry of those large terms. A smaller interaction that lies outside the invariant sector can then supply the first symmetry-allowed contribution to that observable.

We call the resulting structural statement the δJ_7 principle. Its general form is vectorial. The lower-symmetry Hamiltonian can contain several coordinates in the non-invariant operator sector, and symmetry constrains their collective transformation rather than guaranteeing oddness in any one coordinate while the others are held arbitrarily fixed. The familiar one-parameter relation is therefore a controlled slice of a more general source-sector statement. No divergence proportional to an inverse weak-coupling scale follows, and no material-independent hierarchy is implied.

For MnF_2 , the natural source coordinate within the range-seven isotropic exchange description is the difference between the two crystallographically distinct seventh-neighbor bond families. The present paper formalizes that source-sector statement in operator space, fixes the otherwise dangerous bond-label sign convention, and makes explicit the conditions under which the scalar δJ_7 parity series is valid. A companion analysis, *Symmetry-Matched Polarization Analysis in MnF_2 : Separating Spectral Resolution from Chiral Sensitivity in Altermagnetic Neutron Scattering*, develops the complementary readout problem: how polarization projection, domain population, and instrumental transfer determine whether a nonzero antisymmetric response reaches the detector. The present work instead addresses the source side: which Hamiltonian sector can open the response channel, and when a particular coordinate inside that sector can be identified as its leading source.

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II. A CONDITIONAL SYMMETRY COMPLEMENT IN OPERATOR SPACE

The operator decomposition used here is canonical only after the reference problem has been specified. Let $\mathcal{V}$ be a finite-dimensional real vector space spanned by a declared set of Hermitian interaction operators—for example, local terms within a fixed interaction range and unit cell. Let G_0 be an exact symmetry group of a declared reference Hamiltonian H_0 , represented by unitary operators U_g . For a finite group define the Reynolds average

$$\mathcal{P}_{G_0}[X] = \frac{1}{|G_0|} \sum_{g \in G_0} U_g X U_g^{-1}, \quad (1)$$

and for a compact continuous group replace the sum by the normalized Haar integral. The map is idempotent,

$$\mathcal{P}_{G_0}^2 = \mathcal{P}_{G_0}, \quad (2)$$

so

$$\mathcal{V} = \text{im } \mathcal{P}_{G_0} \oplus \ker \mathcal{P}_{G_0}. \quad (3)$$

With a G_0 -invariant inner product this decomposition is orthogonal. On a finite Hilbert space one may use the normalized Hilbert–Schmidt product, $(X, Y) = \text{Tr}(XY)/\dim \mathcal{H}$. For an extended lattice the present construction is understood on the chosen local/operator-coefficient space or with a normalized trace density; no unnormalized Hilbert–Schmidt norm of an extensive thermodynamic operator is required.

The first term in Eq. (3) is the G_0 -invariant sector. We call the second the *Hamiltonian symmetry complement*. This definition is conditional on the chosen reference group, representation, and operator space. Group averaging gives a unique projector once those ingredients are fixed; the physical choice of G_0 and of the retained operator basis is a modeling decision and is not made canonical by the projector itself. In particular, a restricted Hamiltonian can possess an emergent or spuriously enhanced symmetry that disappears when additional symmetry-allowed interactions are retained.[7]

Choose a basis $\mathbf{V} = (V_1, \dots, V_m)^T$ for the relevant complement block and write

$$H(\boldsymbol{\lambda}) = H_0 + \boldsymbol{\lambda}^T \mathbf{V}, \quad \boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_m)^T. \quad (4)$$

For $s \in G_0$, let

$$U_s \mathbf{V} U_s^{-1} = R_s \mathbf{V}. \quad (5)$$

Then the coefficient vector transforms as

$$\boldsymbol{\lambda} \mapsto \rho_s \boldsymbol{\lambda}, \quad \rho_s \equiv R_s^T, \quad (6)$$

and therefore

$$U_s H(\boldsymbol{\lambda}) U_s^{-1} = H(\rho_s \boldsymbol{\lambda}). \quad (7)$$

The scalar family $H(\boldsymbol{\lambda}) = H_0 + \lambda V_-$ is the special case $m = 1$ with $\rho_s = -1$.

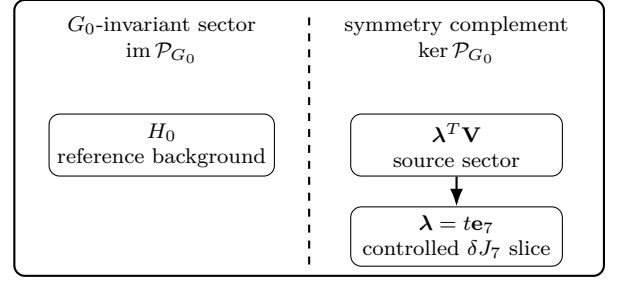


FIG. 1. Operator-space form of the hardened δJ_7 principle. A specified reference Hamiltonian lies in the invariant sector, while a generally multidimensional source vector occupies the symmetry complement. The scalar δJ_7 argument is recovered only on a controlled one-dimensional slice of that sector. The schematic was rendered from deterministic TikZ code developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the mathematical content and labels were specified and verified by the author.

III. COVARIANCE AND THE SCALAR PARITY LIMIT

Let an observable O transform under the selected operation as

$$U_s O U_s^{-1} = \chi_O(s) O, \quad \chi_O(s) = \pm 1, \quad (8)$$

and define the equilibrium expectation

$$F_O(\boldsymbol{\lambda}) = \text{Tr}[\rho_{\boldsymbol{\lambda}} O], \quad (9)$$

with $\rho_{\boldsymbol{\lambda}} = f[H(\boldsymbol{\lambda})]/\text{Tr } f[H(\boldsymbol{\lambda})]$ for any equilibrium function f , including the Boltzmann weight. Equation (7) gives the exact covariance relation

$$F_O(\rho_s \boldsymbol{\lambda}) = \chi_O(s) F_O(\boldsymbol{\lambda}). \quad (10)$$

The same statement applies to appropriately paired dynamical response functions and, below a symmetry-breaking transition, to specified symmetry-related domain branches rather than to an uncontrolled domain average.

A particularly transparent case occurs when the selected source block is odd as a whole, $\rho_s = -I$. Then an odd response satisfies

$$F_-(-\boldsymbol{\lambda}) = -F_-(\boldsymbol{\lambda}). \quad (11)$$

If analytic at the reference point, its multivariate expansion contains only monomials of odd total degree,

$$F_-(\boldsymbol{\lambda}) = \mathbf{k}^T \boldsymbol{\lambda} + \sum_{abc} K_{abc} \lambda_a \lambda_b \lambda_c + O(\|\boldsymbol{\lambda}\|^5). \quad (12)$$

Symmetry fixes the allowed transformation character; it does not guarantee that any particular component of $\mathbf{k}$ is nonzero.

The scalar odd-power series follows after a slice is declared. Let $\boldsymbol{\lambda} = t \mathbf{e}_7$ with all other co-transforming odd

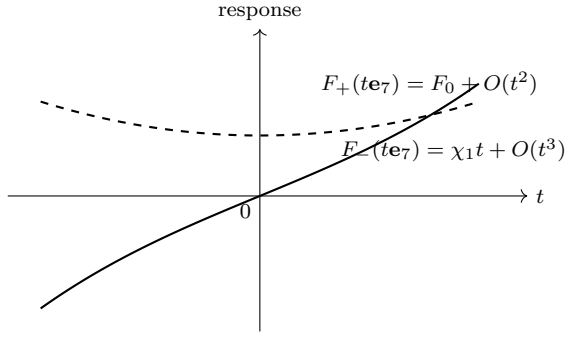


FIG. 2. Parity structure along a controlled one-dimensional source slice. The solid curve is schematic for a symmetry-odd response and the dashed curve for a symmetry-even background. The odd/even power statement applies to this slice; unconstrained additional odd coordinates generally spoil oddness in t alone. The schematic was rendered from deterministic TikZ code developed with substantive assistance from OpenAI ChatGPT (GPT-5.6 Sol); the mathematical content and labels were specified and verified by the author.

coordinates set to zero or otherwise constrained so that the slice is mapped to itself by s . Then

$$F_-(te_7) = \chi_1 t + \chi_3 t^3 + O(t^5), \quad (13)$$

$$F_+(te_7) = F_0 + c_2 t^2 + c_4 t^4 + O(t^6). \quad (14)$$

The phrase “generically linear” means that χ_1 is symmetry allowed and nonzero absent an additional cancellation. Extra symmetry, a matrix-element zero, or a special momentum can force $\chi_1 = 0$.

The slice condition is essential. If a second odd coordinate μ is present and the operation sends $(\lambda, \mu) \mapsto (-\lambda, -\mu)$, the exact statement is

$$F_-(-\lambda, -\mu) = -F_-(\lambda, \mu), \quad (15)$$

not, in general, $F_-(-\lambda, \mu) = -F_-(\lambda, \mu)$. Terms such as $\lambda^2 \mu$ and $\lambda \mu^2$ are allowed because their total degree is odd. Consequently, oddness in δJ_7 alone is a model-and-slice statement, not a theorem in the presence of unconstrained co-transforming source coordinates.

If $F_0 \neq 0$, a normalized contrast along the controlled slice behaves as

$$\mathcal{A}(t) = \frac{F_-(te_7)}{F_+(te_7)} = \frac{\chi_1}{F_0} t + O(t^3). \quad (16)$$

There is no generic $1/t$ divergence. A large measured percentage can arise from a large response coefficient, a small even reference channel, favorable geometry, domain preparation, or engineered transduction. These are properties of the response and measurement map, not an inverse-power law attached to the Hamiltonian hierarchy.

IV. δJ_7 AS A SOURCE COORDINATE IN MnF_2

A. Bond-space decomposition

In rutile MnF_2 , the two Mn sublattices are connected by a crystal rotation combined with translation (and, in the magnetic symmetry, time reversal as appropriate), rather than by a simple lattice translation. The fluoride environment makes two seventh-neighbor in-plane exchange paths crystallographically distinct.[6, 10]

For sign-sensitive statements we fix the bond convention used by Jiang *et al.*: on sublattice 1, J_{7a} labels the in-plane direction $(1, -1, 0)$ and J_{7b} the direction $(1, 1, 0)$, with the latter path bridged by two fluorine ions; the directions are interchanged on sublattice 2.[10] Define

$$\bar{J}_7 = \frac{J_{7a} + J_{7b}}{2}, \quad \delta J_7 \equiv J_{7b} - J_{7a}. \quad (17)$$

Let

$$B_a = \sum_{\langle ij \rangle \in 7a} \mathbf{S}_i \cdot \mathbf{S}_j, \quad B_b = \sum_{\langle ij \rangle \in 7b} \mathbf{S}_i \cdot \mathbf{S}_j. \quad (18)$$

Then the seventh-neighbor isotropic exchange is exactly

$$H_7 = \bar{J}_7(B_a + B_b) + \frac{\delta J_7}{2}(B_b - B_a). \quad (19)$$

The relevant odd/even decomposition is with respect to the *enhanced-symmetry reference model*, not an extra symmetry of the ligand-decorated rutile crystal. Faure *et al.* note that when the fluoride decoration is removed to form the Mn-only body-centered-tetragonal reference, the two seventh-neighbor paths become equivalent.[6] Let s_0 denote a reference-model operation that exchanges those path classes. Then $B_a + B_b$ is even under s_0 , whereas $B_b - B_a$ is odd. Thus

$$V_7 \equiv \frac{B_b - B_a}{2} \quad (20)$$

occupies the relevant complement sector of the reference model, while δJ_7 is its signed coordinate once the ligand environment lifts that equivalence. Equation (19) defines the material-specific δJ_7 direction inside the more general source space of Eq. (4). The scalar parity construction is valid when the stated range-seven model is taken as the operator space and all other source coordinates that co-transform under the selected reference operation are absent, constrained by independent information, or reversed together. If an extended Hamiltonian contains additional unconstrained operators with the same transformation character, the chiral response constrains a linear combination of those coordinates rather than δJ_7 alone.

This is also why bond labels must not be compared by table order alone. Jiang *et al.* explicitly note that the J_{7a}/J_{7b} labels in the Faure parametrization are geometrically opposite to their convention.[10] Consequently, $|J_{7a} - J_{7b}|$ is convention-safe for magnitude comparisons, whereas a signed δJ_7 requires a fixed bond geometry, domain, momentum, and field convention.

B. Magnitude hierarchy is not a gain law

Faure *et al.* report, in their own labeling, a representative fit with $J_2 = 0.287(3)$ meV, $J_{7a} = -0.006(3)$ meV, and $J_{7b} = -0.002(3)$ meV, corresponding to a seventh-neighbor difference of about 4 μeV in magnitude.[6] The ratio

$$\eta_7^{\text{fit}} \equiv \frac{|J_{7a} - J_{7b}|}{|J_2|} \approx 1.4 \times 10^{-2} \approx \frac{1}{72} \quad (21)$$

is useful bookkeeping for that parametrization, but it is not an observable gain.

A later first-principles total-energy mapping by Jiang *et al.*, using the explicit geometry convention of Eq. (17), gives $\delta J_7 \simeq +7.96$ μeV and $J_2 \simeq 338.9$ μeV for their production parameter set, a ratio of about 2.35×10^{-2} . [10] The difference between these numerical ratios reinforces the correct lesson: the hierarchy is model- and parametrization-dependent, whereas the transferable statement is the transformation character of the source coordinate.

C. Why this coordinate is special in the current spin-wave model

The polarized-neutron analysis of Faure *et al.* examined the symmetry-allowed two-spin anisotropic interactions through seventh neighbor that enter quadratic linear spin-wave theory. Their supplemental calculation concludes that, within that model class, the imbalance between the Heisenberg J_7 couplings is the sole two-spin coupling that generates a nonvanishing chiral term in the neutron cross section.[6] This is a strong but deliberately scoped statement. It does not prove that δJ_7 is the unique source in every extended Hamiltonian: longer-range, multispin, or higher-order perturbative terms could occupy the same symmetry sector.

The source coordinate also exists in the nonrelativistic problem. Altermagnetism is defined in the zero-spin-orbit-coupling limit, and Jiang *et al.* extract the isotropic exchange imbalance from non-SOC total-energy mapping before treating small relativistic anisotropies separately.[10] Spin-orbit coupling may modify real-material responses, and spin-point-group analyses distinguish ideal zero-SOC constraints from those of the actual magnetic point group,[8] but SOC is not required to define the basic δJ_7 source term.

V. FROM THE SOURCE SECTOR TO THE CHIRAL RESPONSE

Polarized-neutron theory identifies magnon chirality as a symmetry-selective channel.[4] In the vector formulation, the leading source dependence of a chiral correlator near the reference point has the form

$$C(\mathbf{Q}, \omega; \boldsymbol{\lambda}) = \mathbf{K}_C(\mathbf{Q}, \omega)^T \boldsymbol{\lambda} + O(\|\boldsymbol{\lambda}\|^3), \quad (22)$$

TABLE I. Distinct objects that should not be collapsed into one scalar “altermagnetic scale” in MnF_2 . Representative statements are model or experiment dependent.

Object	Role	Representative statement
δJ_7	Hamiltonian source	few μeV ; signed only after fixing bond convention
ΔE_{dip}	coordinate spectral fine structure	up to ~ 178 μeV ; predominantly dipolar in HYSPEC data
$C(\mathbf{Q}, \omega)$	dynamical chiral response	symmetry-odd correlation generated by the full mode structure
I_χ^{obs}	recorded polarized-neutron channel	depends additionally on probe geometry, domains, and instrumental transfer
magnetic multipole	static/order-parameter descriptor	symmetry-related to altermagnetism but not identical to δJ_7 or I_χ

for a source block that is odd as a whole under the selected operation. The response vector $\mathbf{K}_C$ is not universal: it contains mode eigenvectors, matrix elements, dipolar mixing, momentum dependence, and the other Hamiltonian parameters. A component of $\mathbf{K}_C$ can vanish at special momenta even when the corresponding microscopic coefficient is nonzero.

On the controlled MnF_2 slice $\boldsymbol{\lambda} = \delta J_7 \mathbf{e}_7$ used in the stated range-seven source model, Eq. (22) reduces to

$$C(\mathbf{Q}, \omega; \delta J_7) = \mathcal{K}_{C,7}(\mathbf{Q}, \omega) \delta J_7 + O(\delta J_7^3). \quad (23)$$

This equation is therefore conditional: it is not a claim that every symmetry-allowed extension of the MnF_2 Hamiltonian has only one odd source coordinate. More generally,

$$C = \mathcal{K}_{C,7} \delta J_7 + \sum_{a \neq 7} \mathcal{K}_{C,a} \lambda_a + O(\|\boldsymbol{\lambda}\|^3), \quad (24)$$

so a single chiral observable identifies δJ_7 uniquely only when the competing coordinates are independently constrained or additional readouts make the source vector identifiable.

The companion readout analysis can then be summarized at series level, on the same controlled slice, as

$$I_\chi^{\text{obs}}(\mathbf{Q}, \omega) = \mathcal{T}_R [\eta_D \mathcal{K}_{\chi,7}(\mathbf{Q}, \omega) \delta J_7 + O(\delta J_7^3)], \quad (25)$$

where η_D is the signed antiferromagnetic-domain imbalance, $\mathcal{K}_{\chi,7}$ absorbs the model-specific chiral response and neutron-polarization geometry, and $\mathcal{T}_R$ represents instrumental transfer. Equation (25) is a structural bridge, not an independently fitted MnF_2 formula. It makes explicit why neither $|\delta J_7|/|J_2|$ nor energy resolution alone predicts the measured polarization asymmetry.

The experimental hierarchy is consistent with this separation. Faure *et al.* report a maximum doublet splitting

of about 178 μeV dominated by long-range dipolar coupling, while the polarization-odd chiral fraction reaches roughly 5–11% on selected momentum cuts in a domain-biased sample.[6] Those quantities are not different units of the same scalar. The first is primarily a spectral mixing/splitting scale; the second is a projected dynamical response whose source-side symmetry dependence can be controlled by a much smaller exchange imbalance.

Jiang *et al.* provide an independent illustration of transduction rather than inverse amplification. Their two-mode analysis gives a signed exchange detuning that is odd between rotation-related momenta while the dipolar mixing is common to the pair; at zero field the exchange contribution enters the splitting in quadrature and loses its sign, whereas a longitudinal field shifts the minimum of the splitting linearly with the signed detuning.[10] Their predicted $|B^*| \simeq 0.34$ T and approximately 0.14 meV fixed-field partner contrast are therefore examples of an engineered measurement map. They do not make δJ_7 energetically large. Quantitative optimization of such readouts belongs to the downstream experimental-design problem rather than to the source principle itself.

VI. WHAT THE δJ_7 PRINCIPLE DOES NOT SAY

a. No universal inverse amplification. There is no evidence for a material-independent hiddenness ratio such as $\eta \sim 10^{-2}$, and Eqs. (13)–(16) provide no theorem $\mathcal{A} \propto 1/\eta$. Solov'ev's recent electronic-structure analysis further illustrates that the physical consequence of a small altermagnetic parameter is response dependent: it can remain suppressed in some channels while entering other optical observables much more directly.[11]

b. No identity with a magnetic multipole. δJ_7 is a Hamiltonian coefficient. A magnetic multipole is an order-parameter or magnetization-density descriptor. Bhowal and Spaldin identified a ferroically ordered octupolar description of *d*-wave altermagnetism,[3] while recent polarized-neutron diffraction reports a rank-5 O_{52} component and finite fluorine spin polarization in MnF_2 . [9] These descriptions can be symmetry compatible without being the same physical object. Establishing the probe-to-tensor correspondence is a separate question.

c. No ordering-temperature estimate from $|\delta J_7|/k_B$. The ordered phase is stabilized by the collective exchange network and has $T_N \approx 67$ K.[6] A much smaller coefficient can select or reveal a fine symmetry sector inside that phase. Its microscopic energy scale therefore does not by itself set the temperature at which every symmetry-sensitive observable must disappear.

d. No claim of absolute uniqueness. The strongest supported uniqueness statement is model scoped: in the range-seven, two-spin, quadratic spin-wave analysis of Faure *et al.*, the Heisenberg J_7 imbalance is the sole

coupling found to generate the nonzero chiral term.[6] A more complete Hamiltonian could contain additional operators in the same nontrivial sector. In that case the observable can identify a source-sector combination before it identifies a unique microscopic coordinate.

VII. FALSIFICATION TESTS FOR SOURCE ATTRIBUTION

A source-coordinate claim should be tested by transformation structure and source identifiability, not only by fit quality. Table II states a minimal set of tests for MnF_2 or any material to which the symmetry-complement construction is applied.

These tests distinguish a useful source principle from a numerological hierarchy. In particular, a nonzero δJ_7 is not established merely because a small chiral signal exists, and a chiral signal is not excluded merely because one energy splitting is unresolved. Source-sector identification, coordinate identification, and measurement transmission are separate inference steps.

VIII. DISCUSSION

The hardened δJ_7 principle is a statement about conditional importance in a declared model space. “Dominant” asks which interactions control the largest energy scales. “Functional” asks which source-sector directions control whether a selected response is symmetry allowed. The two rankings need not coincide, and the source sector need not be one dimensional.

MnF_2 makes the distinction unusually transparent. The dominant Heisenberg exchange is hundreds of μeV , the visible finite-momentum doublet can be dominated by a dipolar scale of order $10^2 \mu\text{eV}$, and the source-side seventh-neighbor imbalance is only several μeV . [6, 10] Within the range-seven, two-spin, quadratic spin-wave analysis of Faure *et al.*, the chiral neutron term vanishes for the other examined anisotropic couplings and is opened by the J_7 imbalance.[6] That result makes δJ_7 a distinguished coordinate on that stated model slice. It does not establish that every more complete Hamiltonian has a one-dimensional odd source space.

This distinction also clarifies the role of the fluoride ligands. The Mn moments and dominant exchanges set most of the magnetic energy, while the ligand-decorated rutile geometry encodes the distinction between the two seventh-neighbor paths. Recent diffraction finds finite fluorine spin polarization as well as anisotropic Mn magnetization, underscoring that symmetry-carrying structural degrees of freedom need not be magnetically inert.[9] None of this requires identifying the ligand moment, a magnetic multipole, and δJ_7 as the same quantity.

The vector formulation improves source attribution in two ways. First, it makes degeneracy explicit: a

TABLE II. Falsification tests for assigning a weak Hamiltonian coordinate as the source of a symmetry-selective response.

Test	Required behavior	Failure would imply
Reference null	Set the declared complement sector to zero while retaining the invariant reference Hamiltonian; the targeted odd response must vanish.	A retained term already opens the channel, or the proposed reference symmetry/operator space is incomplete.
Slice control	For a scalar δJ_7 parity claim, identify and suppress, constrain, or jointly reverse every other coordinate that co-transforms with it.	Oddness in δJ_7 alone is not established; the response may depend on a source-sector combination.
Signed reversal	Apply the exact symmetry partner of the declared source vector; the correspondingly odd response must reverse.	The response is not controlled by the assigned representation or contains unresolved competing mechanisms.
Even-channel control	Quantities even under the same operation must obey the corresponding covariance constraint and, on a controlled odd slice, have no symmetry-required linear term.	Leakage between channels, an incorrect parity assignment, or an uncontrolled source coordinate.
Convention audit	Bond, domain, momentum, and field definitions must be stated before comparing signs across calculations or experiments.	Apparent sign disagreement can be a relabeling artifact rather than physics.
Model-completeness test	Add all symmetry-allowed terms at the stated perturbative order and determine which occupy the same source representation.	The source cannot be uniquely identified from the targeted observable alone.
Independent transduction	A second signed readout should constrain the same source vector and infer a compatible δJ_7 on the declared slice.	The original attribution is underconstrained or the response kernel is incomplete.

single odd observable constrains $\mathbf{K}_C^T \boldsymbol{\lambda}$, not necessarily one microscopic coefficient. Second, it makes the route to uniqueness explicit: constrain competing coordinates from independent information or design additional signed readouts whose response vectors span the relevant source subspace. Thus the downstream experimental-design problem is naturally a source-vector identifiability problem rather than a search for a universal weak-coupling scalar.

The transferable content is therefore structural rather than numerical. A candidate system should specify, before using the target response to choose them: a reference Hamiltonian H_0 ; its emergent symmetry G_0 ; the retained operator space; the non-invariant source representation; the candidate coordinate or controlled slice; an observable that vanishes in the reference limit; the response vector including competing perturbations; and a reversal or complementary measurement capable of testing source identifiability. Cross-material generalization should follow such explicit decompositions rather than precede them.

IX. CONCLUSION

We have formulated the δJ_7 principle as an operator-space source-selection statement that remains valid when the symmetry complement is multidimensional. For a specified reference group and operator space, group averaging separates invariant and non-invariant sectors. A source vector $\boldsymbol{\lambda}$ in the latter transforms under a representation ρ_s , and a symmetry-selective observable obeys the exact covariance relation $F_O(\rho_s \boldsymbol{\lambda}) = \chi_O(s) F_O(\boldsymbol{\lambda})$. When

an entire source block changes sign, an odd response contains only terms of odd total degree. The familiar one-parameter odd series follows only on a controlled slice for which other co-transforming odd coordinates are absent or independently constrained.

In MnF_2 , the two seventh-neighbor Heisenberg bond families provide a concrete such direction. With bond geometry fixed explicitly, the seventh-neighbor exchange decomposes into an invariant average and a bond-odd operator whose signed coordinate is $\delta J_7 = J_{7b} - J_{7a}$. Existing polarized-neutron modeling identifies this imbalance as the sole two-spin coupling through seventh neighbor that produces a nonvanishing chiral term at quadratic spin-wave order, while phenomenological and first-principles determinations place its magnitude at only a few μeV . Within that model class, the small coefficient can therefore be the first allowed source of the selected response without becoming a dominant Hamiltonian energy.

The practical consequence is a sharper separation of claims. δJ_7 is a Hamiltonian source coordinate, not a magnetic multipole, not the dipolar magnon splitting, and not the recorded neutron asymmetry. A measured odd response can identify δJ_7 only conditional on the declared source slice or on sufficient complementary data to resolve competing coordinates. This vector source-response formulation turns “weak but important” into a falsifiable symmetry-and-identifiability statement and supplies the appropriate starting point for signed readout engineering and cross-material tests.

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DATA AVAILABILITY

This theoretical analysis generated no new experimental datasets and did not independently refit raw neutron-scattering data. Numerical values and experimental observations used for illustration are taken from the cited public literature and associated public records.

CONFLICT OF INTEREST

The author declares no competing financial interest related to this manuscript.

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